

Multiple Choice (10 marks)

- Which of the following is an n-type semiconductor?
 - Silicon
 - Diamond
 - Silicon carbide
 - Arsenic-doped silicon
- Calculate the polarization of a proton in a magnetic field of 335 mT and 10.5 T at 298 K.
 - 6.345×10^{-4} at 0.335 T; 9.871×10^{-5} at 10.5 T
 - 0.793×10^{-4} at 0.335 T; 6.931×10^{-7} at 10.5 T
 - 1.148×10^{-6} at 0.335 T; 3.598×10^{-5} at 10.5 T
 - 4.126×10^{-3} at 0.335 T; 2.142×10^{-6} at 10.5 T
- Which of the following is lower for argon than for neon?
 - Melting point
 - Boiling point
 - Polarizability
 - First ionization energy
- Of the following atoms, which has the lowest electron affinity?
 - F
 - Si
 - O
 - Ca
- Of the following compounds, which has the lowest melting point?
 - HCl
 - AgCl
 - CaCl₂
 - CCl₄

True / False (10 marks)

Answer True or False

6. True or False: The balanced equation shows a MnO₄⁻ to I⁻ ratio of 6:5.
7. True or False: Nuclear magnetic resonance spectroscopy is a light-scattering technique that analyzes molecular structures.
8. True or False: CH₃F has a ¹H resonance that is closer to TMS than CH₃Cl on a spectrometer with a 12.0 T magnet.
9. True or False: Protons in a molecule with a rotational correlation time of 1 ns have the fastest spin-lattice relaxation at a magnetic field of 3.74 T.
10. True or False: The ⁵⁵Mn nuclide has five allowed energy levels for its nuclear states.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the ratio of line intensities in the EPR spectrum of the t-Bu radical (CH₃)₃C• and explain the significance of these ratios in understanding radical behavior.
12. Discuss the significance of NMR frequency in relation to nuclear spin properties, and explain why ¹⁷O exhibits a frequency of 115.5 MHz in a 20.0 T magnetic field.

End of Paper